DO NOW — RECALL

Lesson 134 · Database Foundations: Flat Files, Tables a

Name: _____ Class: _____ Date: _____

1. What does a cryptographic digest help verify?

2. Which is removed first from a FIFO queue?

3. What is a composite key?

4. What are the outputs of a half adder when A=1 and B=1?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Integrity, when compared against a trusted expected digest.

2. Model answer

The longest-waiting item at the front.

3. Model answer

A key made from two or more attributes together.

4. Model answer

Sum 0, carry 1.

By the end, you can…

• Duplicating pupil details on each enrolment creates repetition.

• Pupil and Enrolment records separate into related entities.

• Primary key identifies a row; foreign key preserves relation to parent.

Vocabulary — know these terms

record — A row representing one entity instance.

primary key — Unique, non-null identifier for a table row.

foreign key — Column referencing a key in another table.

flat file — Data kept in one simple table or file without relational links.

NEW LEARNING

Read, annotate and explain

Single flat file

Duplicating pupil details on each enrolment creates repetition.

Relational tables

Pupil and Enrolment records separate into related entities.

Keys connect

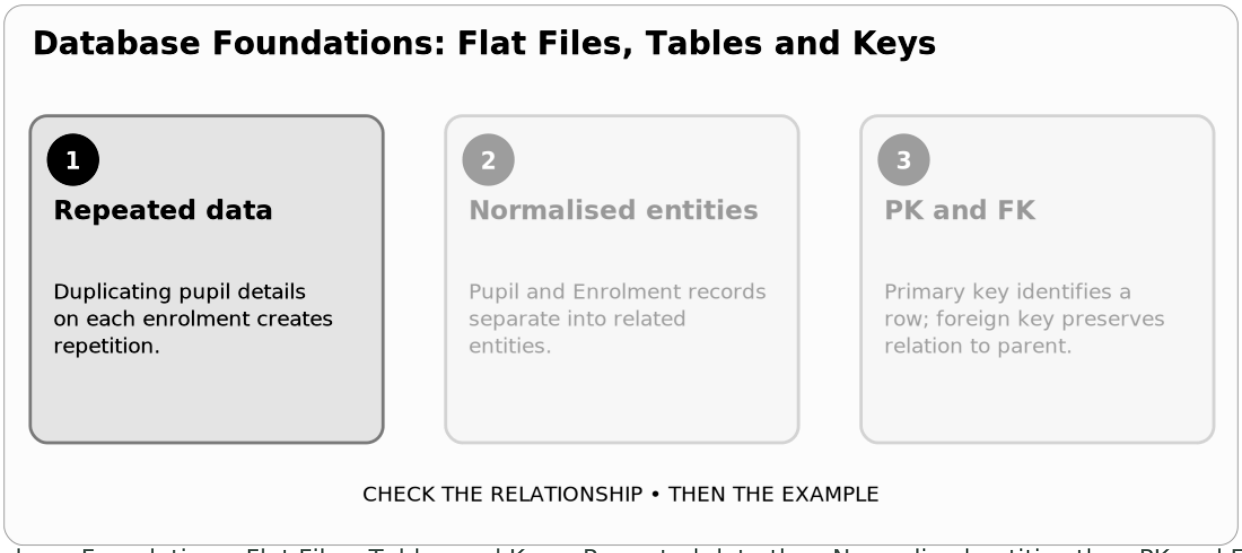
Primary key identifies a row; foreign key preserves relation to parent.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

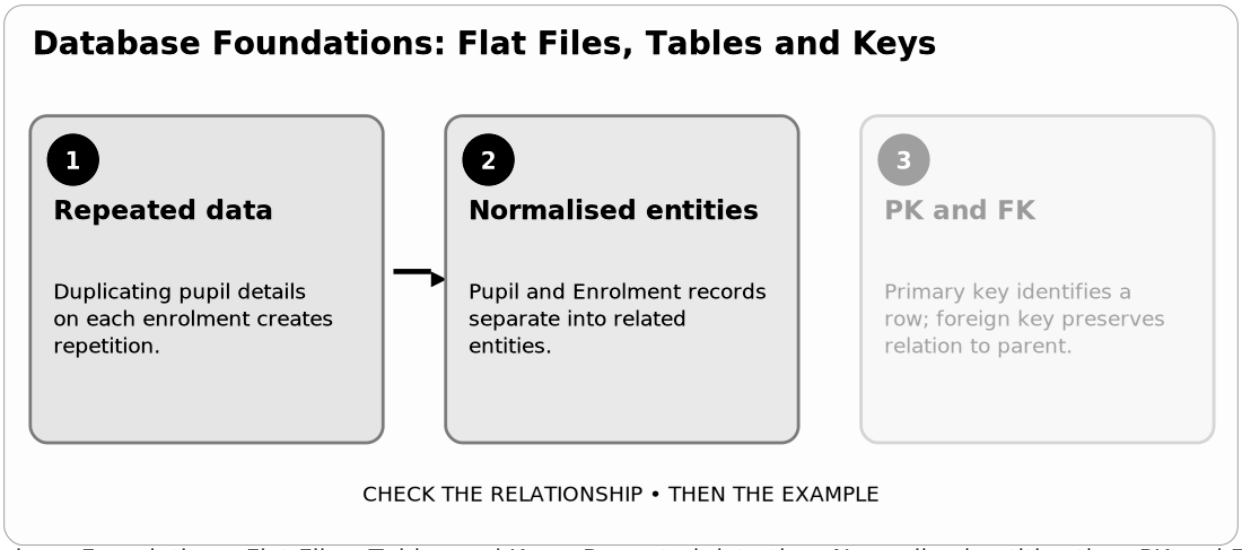
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Database Foundations: Flat Files, Tables and Keys; Repeated data then Normalised entities then PK and FK

New Learning

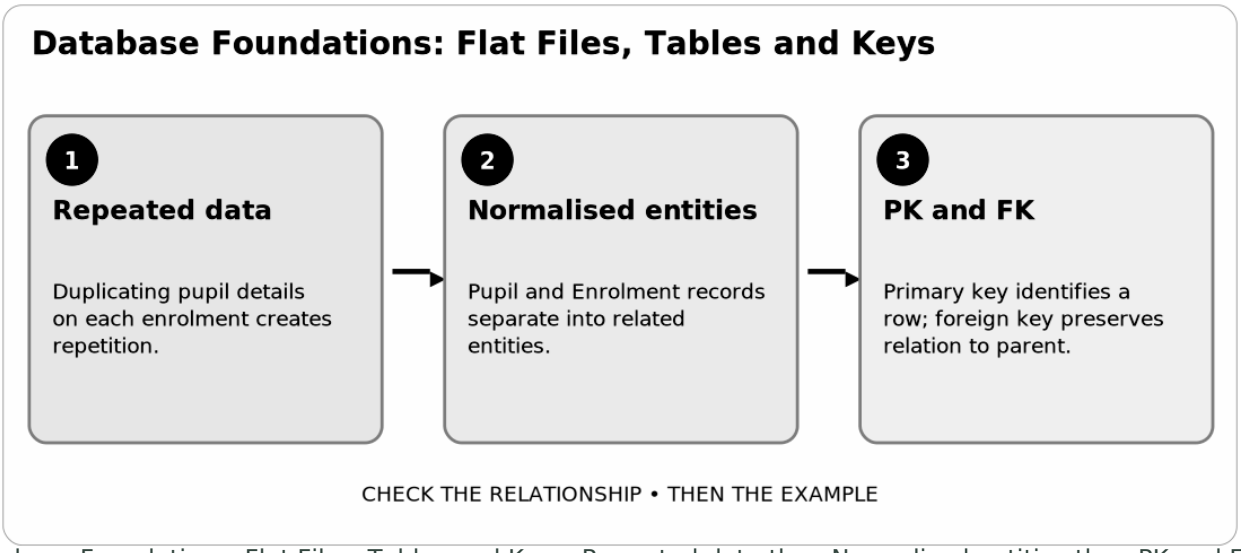


Database Foundations: Flat Files, Tables and Keys; Repeated data then Normalised entities then PK and FK

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Database Foundations: Flat Files, Tables and Keys; Repeated data then Normalised entities then PK and FK

I DO — WORKED EXAMPLE

Follow the worked method

A pupil books several clubs. Why store pupils and club bookings in separate tables?

Pupil ID 7: Sam

Find the given data and name the required result.

Bookings refer to PupilID 7

Explain why this intermediate stage is necessary.

Change name once

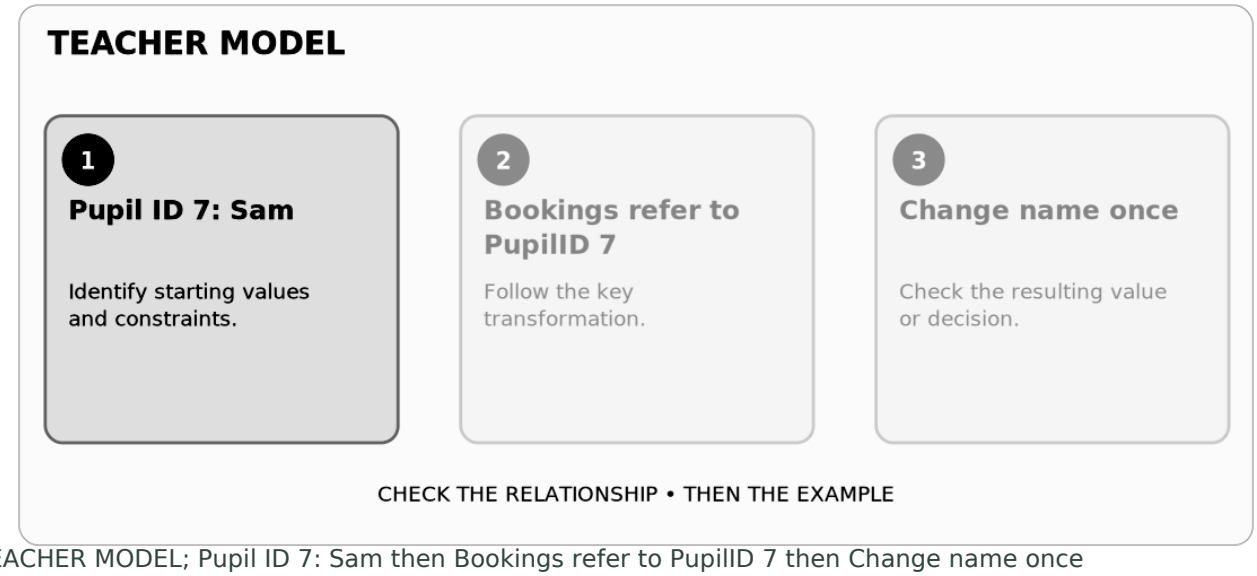
Pupil(PupilID PK, Name) and Booking(BookingID PK, PupilID FK, ClubID FK) separate data. A pupil name is held once and each booking refers to the pupil by ID, reducing update anomalies.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil books several clubs. Why store pupils and club bookings in separate tables?

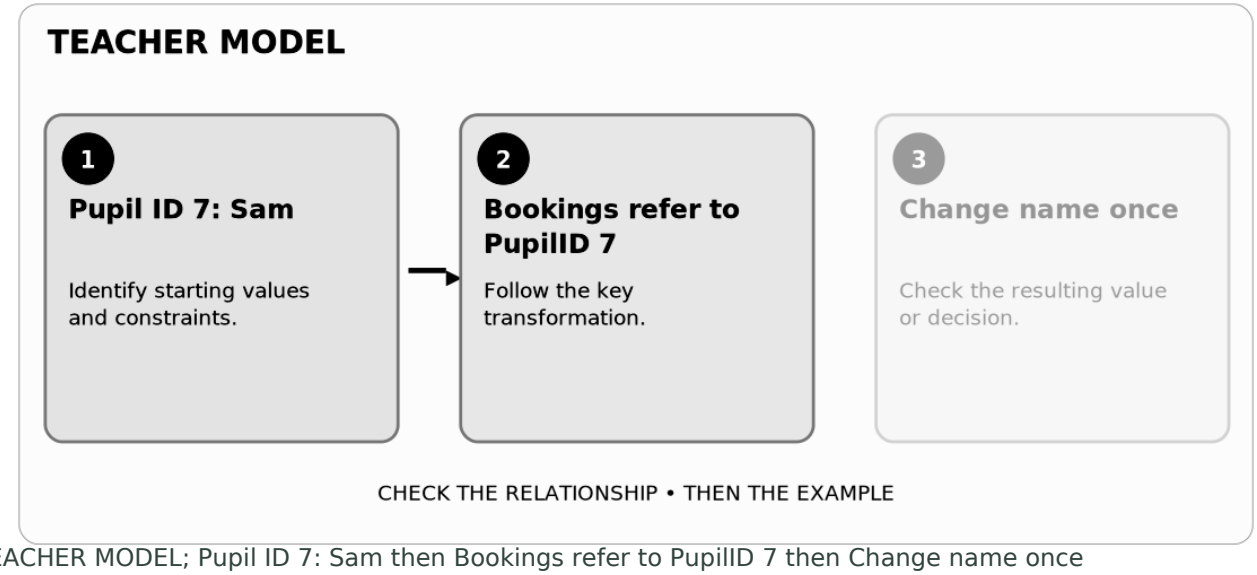
Find the given data and name the required result.



TEACHER MODEL; Pupil ID 7: Sam then Bookings refer to PupilID 7 then Change name once

A pupil books several clubs. Why store pupils and club bookings in separate tables?

Find the given data and name the required result.



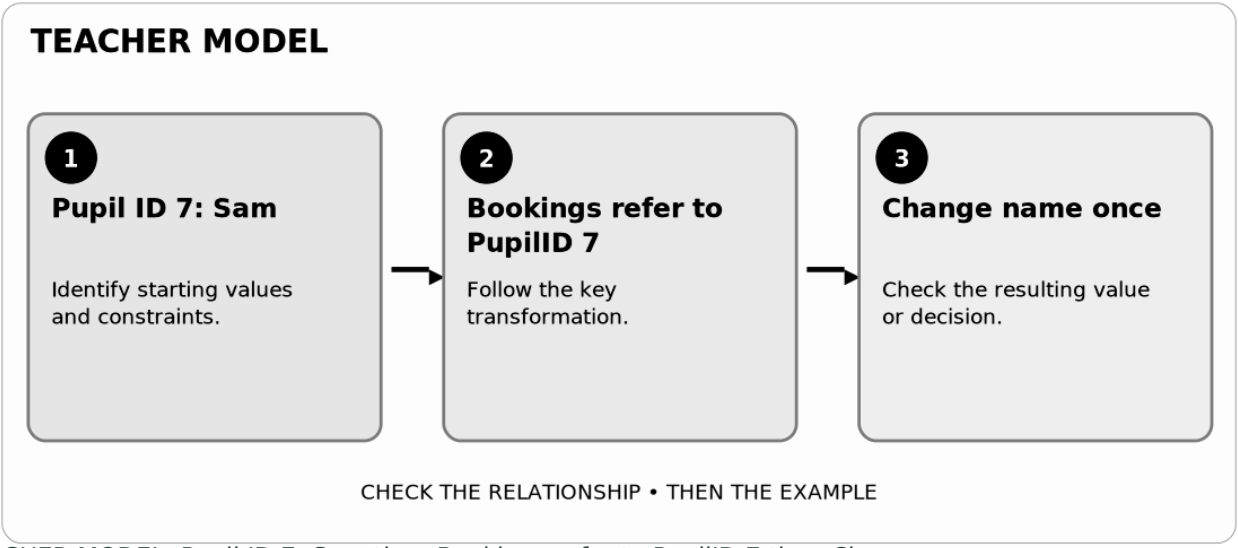
TEACHER MODEL; Pupil ID 7: Sam then Bookings refer to PupilID 7 then Change name once

I Do · original lesson diagrams

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A pupil books several clubs. Why store pupils and club bookings in separate tables?

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TEACHER MODEL; Pupil ID 7: Sam then Bookings refer to PupilID 7 then Change name once

WE DO — GUIDED PRACTICE

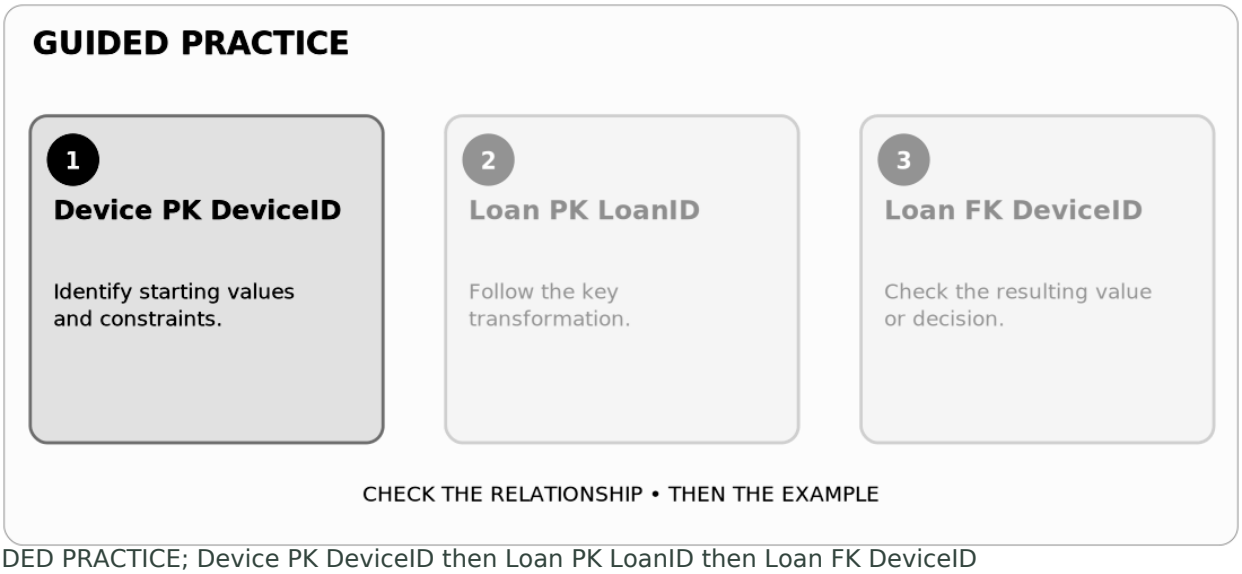
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school has Device(DeviceID, Model) and Loan(LoanID, DeviceID, Borrower). Identify suitable keys.

Find the given data and name the required result.



GUIDED PRACTICE; Device PK DeviceID then Loan PK LoanID then Loan FK DeviceID

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Device PK DeviceID Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define primary key and foreign key using pupil/club tables. [6]

Question 2

Explain two problems with keeping pupil contact details on every booking row. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design a relational structure for Customers placing multiple Orders, showing suitable PKs and FK. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A foreign key value must always be unique in its table.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

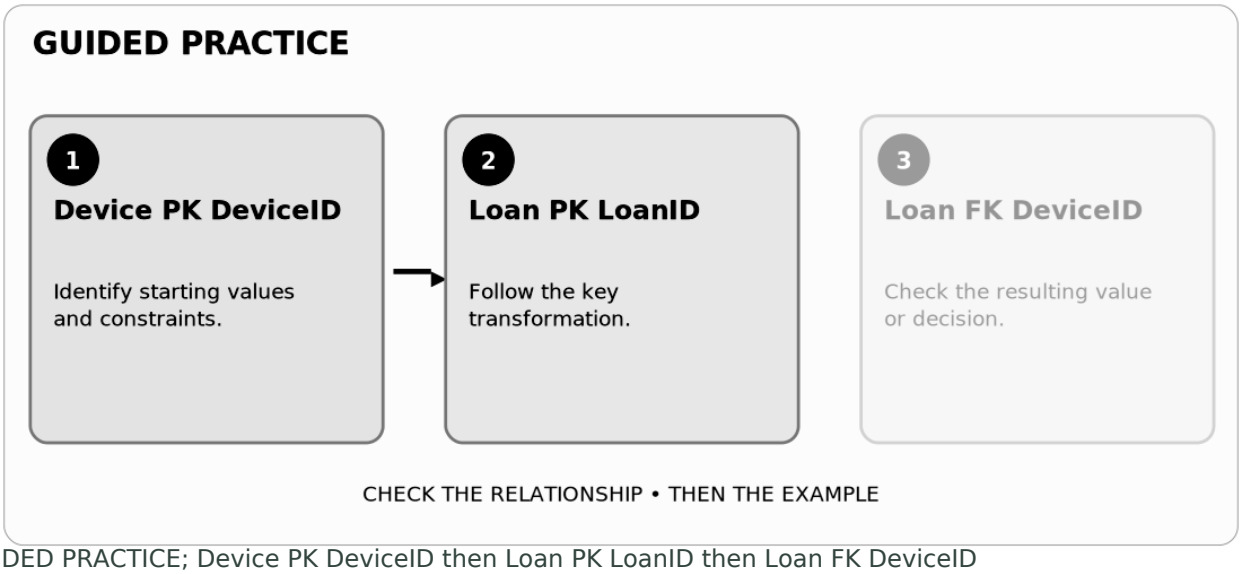
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school has Device(DeviceID, Model) and Loan(LoanID, DeviceID, Borrower). Identify suitable keys.

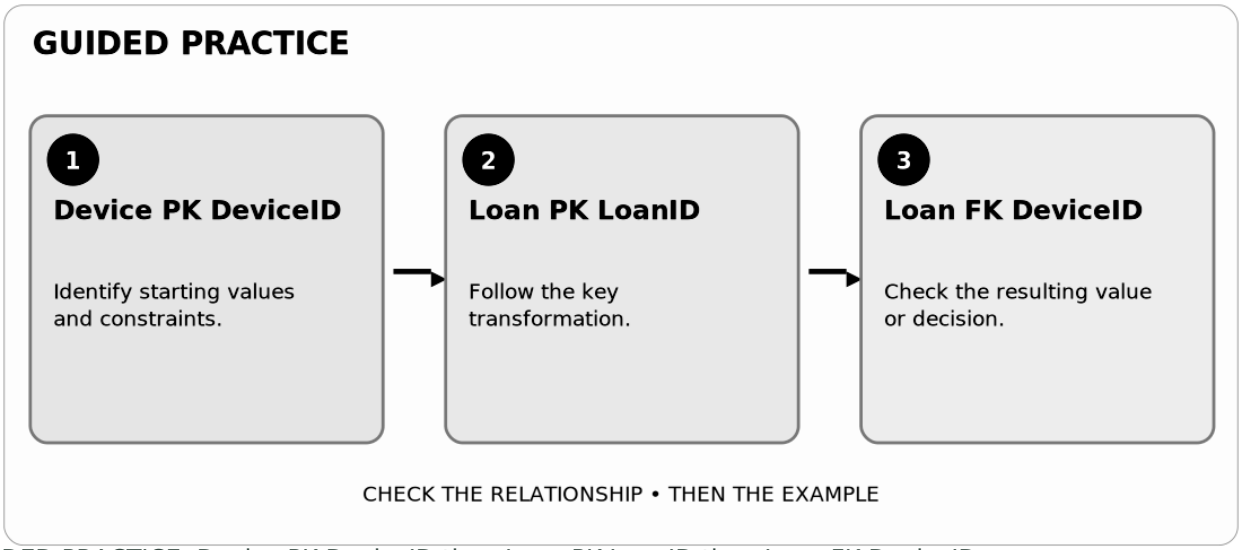
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GUIDED PRACTICE; Device PK DeviceID then Loan PK LoanID then Loan FK DeviceID

A school has Device(DeviceID, Model) and Loan(LoanID, DeviceID, Borrower). Identify suitable keys.

Find the given data and name the required result.



GUIDED PRACTICE; Device PK DeviceID then Loan PK LoanID then Loan FK DeviceID

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Loan PK LoanID

Explain why this intermediate stage is necessary.

Loan FK DeviceID

DeviceID uniquely identifies each device (PK). LoanID identifies each loan (PK). Loan.DeviceID is an FK to Device.DeviceID. Borrower may later reference a Borrower table.

IMPROVE: Explain one correction to your own reasoning.